

Study Guide for Exam #1 with Practice Problems

MATH 241 — Probability, Fall 2026

General Information

The first exam will be held in-class on Tuesday, September 29th. Keep in mind the following things:

- You will have 75 minutes to complete the exam, unless you have an accomodation.
- This is a **closed-book** exam, with the exception of a single 8.5x11" sheet of paper of notes. You can use both sides of the sheet to write down anything.
- You will not need to use any R code on this exam. However, you will want to study methodically how to complete tasks like a Monte Carlo simulation, a random sample with or without weights/replacement, and using functions.
- Cell-phones, laptops, calculators, and any other electronic devices are not permitted. No interaction with anyone is allowed.

Expect the exam to contain about 4-5 questions, with a few subparts to each question.

Topics

0. (A.1): Set notation, including the use of unions, intersections, and complements
1. (Ch. 1: 1.1-1.4, 1.6): Probability and counting
 - Sample spaces, outcomes, and events
 - Constructing new events from the unions, intersections, and complements of other events
 - Probability functions and the axioms of probability
 - Additional properties: complements, generalized additive property, and inclusion-exclusion
 - Combinations, permutations, and the multiplication principle
2. (Ch. 2: 2.1-2.6): Conditional probability
 - Definition and its properties
 - Use of Law of Total Property, Multiplication Rule, and Bayes' Theorem
 - Independence

Tips

Ultimately, practice makes perfect, and especially so in probability. The textbook has many problems that have available solutions online via <https://stat110.net>.

When working on the problems, take the time to write down everything you know about the situation at hand from the problem: events, probabilities, sets, etc. With this information, you may find that some form of a diagram will help here, either as a Venn diagram, a tree diagram, or through counting. You also want to consider what probability you are trying to find, and which formulas/definitions will allow you to use the given information to solve.

Dr. Belton, who has taught the course before, also suggests thinking about the following questions:

- What language in a problem is intended to indicate event unions, event intersections, or event complements? Conditional probability? Independence? Sampling with (or without) replacement? Order mattering or not mattering?
- What relevant theorems, properties, and problem-solving strategies have we seen for finding the probability of event unions, e.g. $P(A \cup B)$?
- What about for finding the probability of event intersections, e.g. $P(A \cap B)$?
- What relevant theorems, properties, and problem-solving strategies have we seen for finding and relating conditional probabilities, e.g. $P(A|B)$?
- What relevant theorems, properties, and problem-solving strategies have we seen for finding unconditional probabilities, e.g. $P(A)$?
- What are some approaches that we've used frequently in this class (e.g. working with complements, identifying partitions) and when do we tend to use them?
- Are there pictures, diagrams, or simple test cases that we can use to build intuition for particular types of problems?

Lastly, make the most of the resources you have. I will have three different office hour time blocks throughout the next week, and we will have a review session on Thursday. Come to Thursday's class with any clarifying questions you would like answered.

Practice Problems

1. For each of the following situations, provide (1) the random experiment that is occurring, (2) the sample space of that experiment, (3) the event we are wanting the probability for, and (4) the value of that probability.
 - (a) We roll five dice altogether, trying to identify the chance we get the same value for all five dice.
 - (b) A tennis match starts with a racket spin (equivalent to a coin flip), with the winner choosing if they want to receive or serve, or pick which side to start from. We want to identify the chances of player 1 choosing the opportunity to serve first. Assume if player 2 wins the spin, he has an equal chance of picking one of the four choices.
 - (c) There is a 3% chance that my bicycle might fail on a single ride for a variety of reasons. We want to determine the chance my bicycle will be okay for the next 10 rides.
2. Suppose Jamie wants to buy 4 new plants for her apartment, but is currently indecisive about which kinds of plants she wants. She visits a nursery and finds that they have 10 monstera (3 of which are small, 4 medium, and 3 large), 3 palms (1 small, 2 large), and 6 prayer plants (5 small, 1 medium). Find the number of possibilities Jamie can buy:
 - (a) 2 small, 1 medium, and 1 large plant
 - (b) At least 3 monstera
 - (c) One of each type of plant (ignoring size)
3. Jack creates a new card game called “Three-Draw”, in which certain hands will win you money. To win money, you can draw the following:
 - All of the same number/face value
 - Cards in increasing or decreasing sequential order (assume $A = 1$, $J = 11$, $Q = 12$, $K = 13$)
 - All of the same suit

Find the probability of getting one of these hands in this game.

4. Determine if the following situations constitute a probability distribution and describe its sample space:
 - (a) Claude notes that for a sample space of $\{0, 1, 2\}$, we have probabilities of $P(\{0\}) = 0.4$, $P(\{1\}) = 0.6$, $P(\{2\}) = 1 - P(\{0\})$.
 - (b) ESPN notes that at the beginning of the 4th quarter of a WNBA game between the Sparks and Mercury, the chance of the Sparks winning is 62% and them losing is 38%.

- (c) Of the three baseball caps that may contain a ball inside, one cap definitely has the ball while the other two do not.
5. Alex, Brianna, Carl, and Darius all decide to play Monopoly and are about to take their first few turns. We only focus on what happens on the first 10 spaces for the first run-through of the board. Note the following:
- Landing on the first and third spaces guarantee a chance to buy a brown property.
 - Landing on the sixth, eighth, and ninth spaces guarantee a chance to buy a light-blue property.
 - Landing on the other spaces (2nd, 4th, 5th, 7th, and 10th) guarantees another action.

Assume we are using fair dice and all dice rolls are independent of each other. Each dice roll consists of rolling two die. Then, find the following probabilities:

- (a) That Alex lands on at least one blue property within two rolls.
- (b) That Carl lands on the same space on the first roll given Brianna landed on the same space on the first roll.
- (c) Everyone fails to land on a light blue or brown property after one roll.
6. The probability of getting two pairs in a game of poker is 4.75%. Explain how I could verify this empirically.
7. In a small community in New York, a census of cars is taken and the following information is determined:
- $P(\text{black vehicle}) = 0.3$
 - $P(\text{white vehicle}) = 0.2$
 - $P(\text{gray vehicle}) = 0.1$
 - $P(\text{vehicle of any other color}) = 0.4$
 - $P(\text{truck}) = 0.25$
 - $P(\text{truck} \mid \text{black}) = 0.15$
 - $P(\text{truck} \mid \text{not black}) = 0.6$

Find the following probabilities:

- (a) $P(\text{black truck})$
- (b) $P(\text{non-black truck})$
- (c) $P(\text{black} \mid \text{truck})$
- (d) $P(\text{black} \mid \text{not a truck})$
8. Suppose we have the following setup for events A , B , and C :

- $P(A) = 0.4$
- $P(B) = 0.25$
- $P(A \cap B) = 0.1$
- $P(A \cap C) = 0.3$
- $P(C) = 0.5$

Answer the following parts:

- Find $P(A|B)$.
 - Find $P(C|A)$.
 - Say $P(A \cup B \cup C) = 1$. Is this possible? If so, what can we say (or what do we know) about the other sets in this space.
9. I'm playing Wordle and am currently stuck after 3 chances. I have 11 letters left to fill 5 spaces. Assume all combinations of letters are possible as a word.
- What is the number of possibilities of filling these spaces up?
 - I have a feeling (and somehow know) that the second and third spaces have an e in them. What is the probability of getting the correct word on that turn, knowing this information?
 - After using my fourth and fifth chances, I have guessed p, e, e, -, - and have five letters left on the final turn. Assuming there are no more duplicates, how many possibilities are left?
10. Darren and Michael play a game of rock, paper, scissors. With their past history, here are the following probabilities of what they will play:
- $P(\text{Darren plays rock}) = 0.5$
 - $P(\text{Darren plays scissors}) = 0.4$
 - $P(\text{Darren plays paper}) = 0.1$
 - $P(\text{Michael plays rock}) = 0.3$
 - $P(\text{Michael plays scissors}) = 0.4$
 - $P(\text{Michael plays paper}) = 0.3$

Determine who has the best chance of winning a single game. Assume that ties may happen.